5 Numerical Experiments

5.1 Protocol and implementation details

We evaluate two questions: whether a LoRA transfers the Chinese ink-wash domain beyond exact training captions, and whether the adapted 20-call model can be compressed to four or two Transformer calls. The complete validated corpus has 260 image–caption pairs (209 plant, 30 animal, 11 web, and 10 other); the primary teacher and distillation experiments use the coherent 209-image plant subset. Images are represented by cached FP16 VAE latents of shape $4 \times 64 \times 64$, and captions by cached T5-XXL embeddings of shape 300×4096. All reported images are 512×512.

The Style Teacher is PixArt-Sigma-XL-2-512-MS with a rank-16 attention LoRA (13.8M trainable parameters). It is sampled for 20 DPM-Solver++ steps with classifier-free guidance (CFG) 1.5. Distillation uses 627 training prompts (three variants of every plant caption) and two deterministic seeds per prompt, giving 1,254 cached teacher trajectories. The joint rank-16 students continue the style adapter and use an 80% pseudo-Huber trajectory objective plus a 20% clean-latent diffusion anchor. The selected 4-step checkpoint is training step 4,500 from a run extended to 6,000 updates (5×10^{-6} learning rate); the 2-step student is trained for 7,000 updates from that adapter (2×10^{-6} learning rate, 50% on-policy roll-in). Student inference uses CFG 1.0 with no unconditional branch and exactly four or two Transformer calls.

The held-out benchmark contains 30 unseen prompts and four fixed seeds per prompt, or 120 images per model. We report CLIP ViT-B/32 text–image cosine similarity, unbiased squared CMMD between normalized CLIP image features and all 209 real plant images (Gaussian RBF bandwidth 10), median denoising time, and the exact forward-call count. For uncertainty, each seed position defines one replicate containing one image for each of the 30 prompts. Table 1 reports mean $\pm$ sample standard deviation (SD) over these four replicates; thus prompt difficulty is held constant instead of being conflated with seed variation. Latency was measured after warm-up on the same local GPU/software environment. No additional model training was performed for the analyses below.

5.2 Style adaptation and held-out generation

Before distillation, we compared Style Teacher LoRAs of ranks 4, 8, 16, and 32 at every 1,000 updates up to 10,000. The official PixArt model is the reference, using the same prompt, seed, 20-step schedule, and CFG 1.5. For a rewritten palm caption, the best checkpoint (rank 4, step 1,000) improves CLIP from 0.3399 to 0.3561 (+4.7%). For an unseen misty-mountain composition, rank 8 at step 4,000 improves 0.3615 to 0.3720 (+2.9%). Later checkpoints frequently decline, and the best rank changes with the prompt; LoRA therefore transfers the style, but held-out checkpoint selection is necessary and "more training" is not a reliable selection rule.

Figure 1 uses the preferred held-out prompt, "A lone fishing boat crossing a vast river beneath high cliffs, traditional Chinese ink wash painting style, shuimo hua," at seed 10163. The official base is semantically correct but relatively photographic. Teacher B establishes the monochrome brush-and-negative-space distribution, while both distilled students preserve the principal cliffs, river, and boat. The 2-step image is softer, consistent with its lower aggregate CLIP score and larger CMMD.

5.3 Few-step quality, efficiency, and seed variance

Table 1 gives the primary quantitative comparison. The 4-step student reaches CLIP 0.3674, or 100.65% of the teacher, while reducing median latency from 2.871 to 0.480 seconds (5.99×). Its CMMD remains close to the teacher. The 2-step student retains 96.84% of teacher CLIP and is 11.78× faster, but its CMMD of 0.001394 lies only 2.00% below the predefined limit $1.5 \times 0.000949 = 0.001423$. Consequently, four steps are the quality-oriented operating point and two steps the latency-oriented point; the data do not justify claiming exact teacher equivalence.

The variance result qualifies the averages. Across seed replicates, the 4-step CLIP advantage over the teacher is 0.0024, smaller than either model's replicate SD; it is therefore best interpreted as quality preservation, not a statistically established improvement. Conversely, the 2-step mean is 0.0115 below the teacher, about 3.9% relative, and its CMMD has the largest SD. At the individual-prompt level, the root-mean-square within-prompt seed SD is 0.0230, 0.0200, and 0.0213 for teacher,

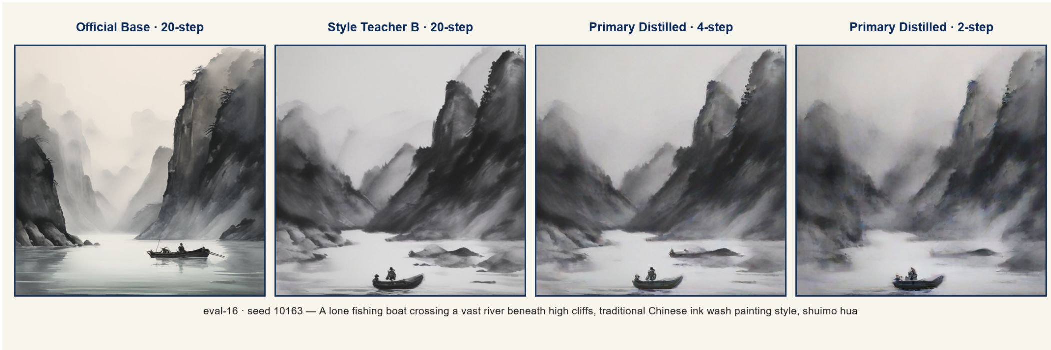


Figure 1: Same unseen prompt and seed for the official 20-step base, 20-step Style Teacher B, primary 4-step student, and primary 2-step student.

Table 1: Primary held-out results. Each value is mean ± SD across four seed replicates (30 unseen prompts per replicate), except Calls and Speedup. CMMD replicate means use 30 generated images against all 209 real plant images.

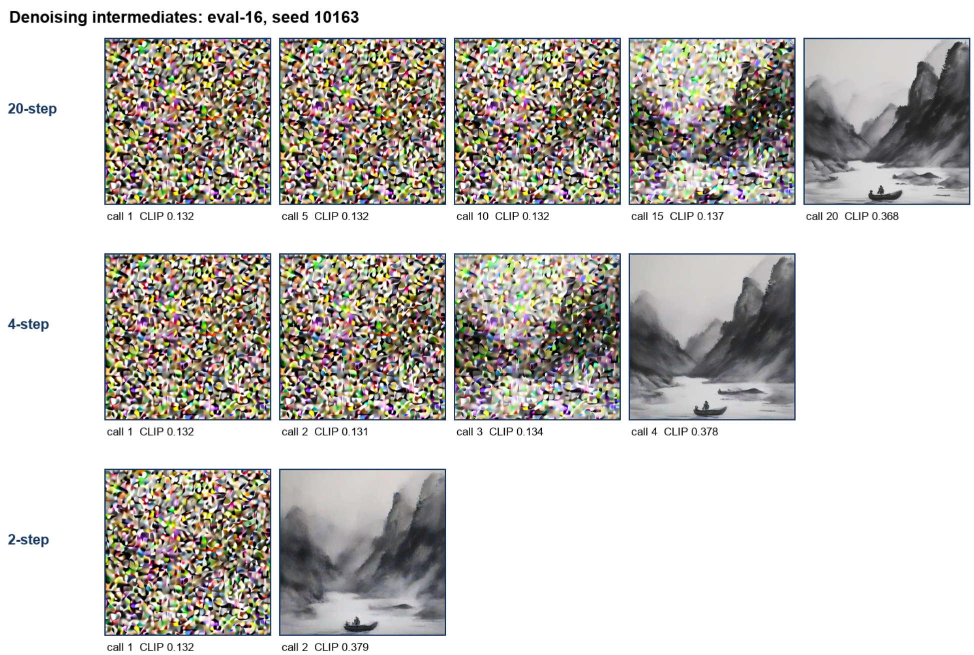
Model	Calls	CLIP ↑	CMMD ↓	Time (s) ↓	Speedup
Teacher B	20	0.3651±0.0040	0.000947±0.000056	2.871±0.006	1.00×
4-step	4	**0.3674±0.0049**	0.001070±0.000047	0.484±0.009	5.99×
2-step	2	0.3535±0.0025	0.001385±0.000102	**0.243±0.010**	**11.78×**

4-step, and 2-step respectively. A single seed is thus insufficient for judging an example even though model-level replicate means are relatively stable.

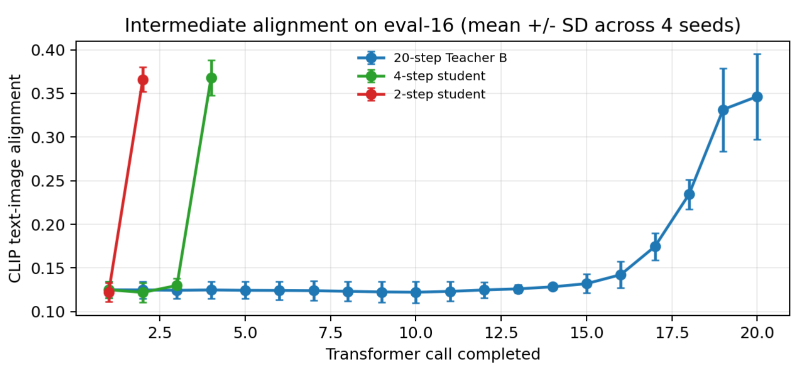
5.4 What happens between the reported endpoints?

The presentation originally compared only completed 20-, 4-, and 2-call generations. To expose the missing intermediate behavior, we reran inference only (no weight updates) on the preferred `eval-16` prompt for seeds 10161–10164. After every Transformer call, we decoded the current latent x_t with the same VAE and computed CLIP. The teacher uses its original 20-step DPM-Solver++ schedule; the students retain their trained deterministic jumps $0 \to 5 \to 10 \to 15 \to 20$ and $0 \to 10 \to 20$.

Figure 2 shows that the teacher's mean CLIP remains near 0.12 through call 14, rises to 0.174 at call 17 and 0.234 at call 18, then reaches 0.346 ± 0.049 at call 20. The 4-step student measures 0.125, 0.122, 0.130, and 0.368 ± 0.020 after calls 1–4; the 2-step student measures 0.122 and 0.366 ± 0.014. Visually, the boat-and-cliff composition likewise emerges in the final jump. This does *not* mean earlier network calls are useless: their decoded x_t is deliberately noisy and may not lie on the natural-image manifold, so CLIP is only a diagnostic of visible emergence rather than a valid per-step training objective. Nevertheless, the result directly verifies that few-step students perform large learned transitions rather than merely stopping the 20-step solver early. All per-call images, per-seed scores, and feature caches are saved for reproducibility.



(a) Decoded intermediate states for seed 10163.



(b) CLIP mean $\pm$ SD over four seeds after every call.

Figure 2: Intermediate-state evaluation for the same held-out fishing-boat prompt. Early decoded x_t states remain noisy; most measurable semantic alignment appears in the final call.